

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

Headline Finding: The 50% task-completion time horizon has doubled approximately every 207 days (~7 months) from 2019–2025.

Paper: “Measuring AI Ability to Complete Long Tasks”

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Audience: AI safety researchers, frontier AI lab leadership, and policy analysts

A New Metric: 50% Task-Completion Time Horizon

1. Build task suite

170 software/research tasks spanning seconds to hours of skilled human effort.

2. Measure humans + AI

Collect human completion times and AI-agent success rates across the same tasks.

3. Fit logistic curve

Estimate duration where success probability = 50%; this is the model's time horizon.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Task Suite Source	Task Count	Human-Time Span	Representative Example Tasks
HCAST	97	1 min–30 hrs	wikipedia_research (1 min); oxdna_simple (9 min); munge_data (56 min)
RE-Bench	7	8 hrs	cuda_backtesting (8 hrs), difficult ML research engineering tasks
SWAA	66	1 sec–30 sec	find_shell_script (3 sec), short single-step software actions

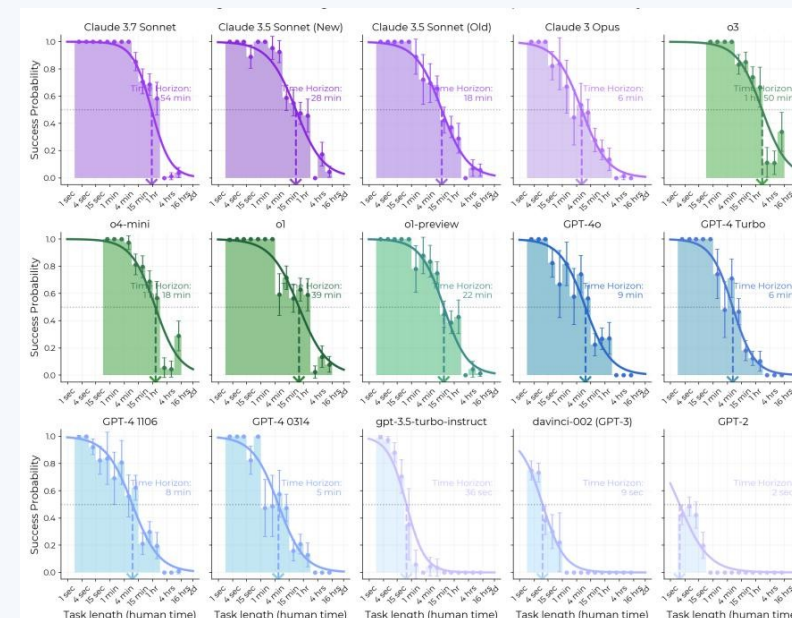
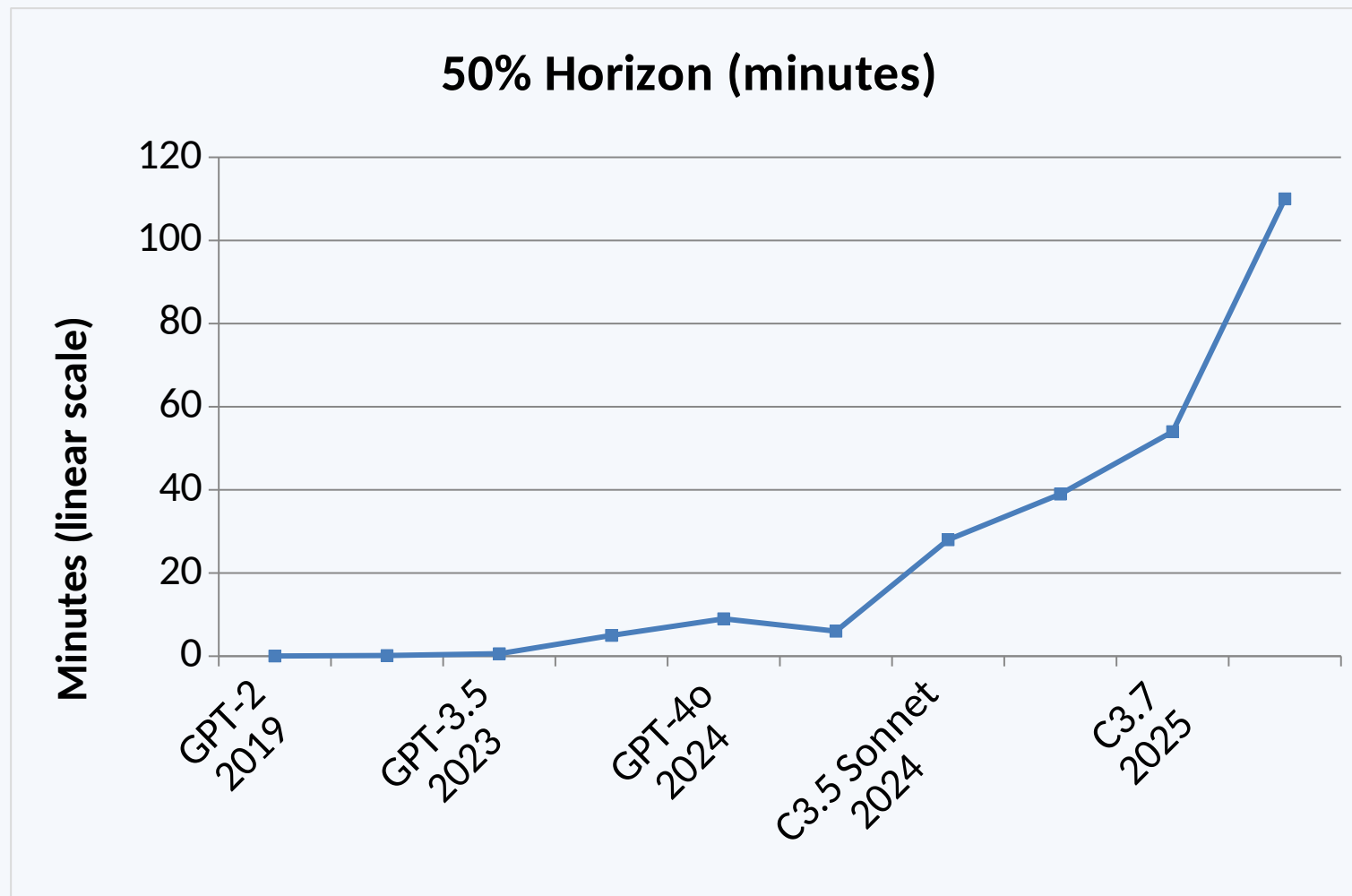
Benchmark coverage and labeling

Total suite: 170 tasks with over 800 human baselines and 2,529 human hours measured.

Design goal: connect very short atomic actions and multi-hour engineering tasks with one common metric.

Task diversity supports cross-model comparison from GPT-2-era systems to frontier reasoning agents.

Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years



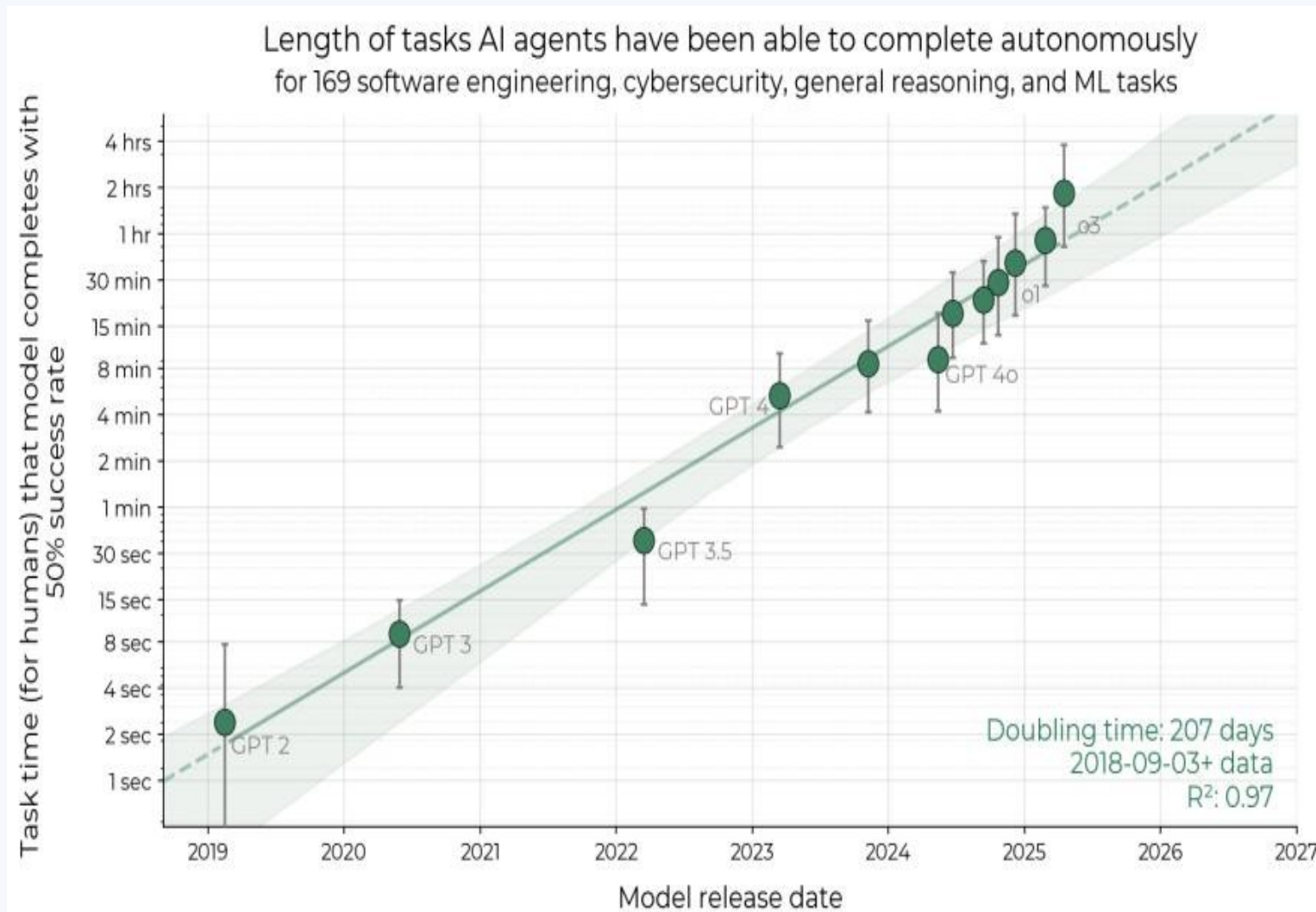
Trend fit summary

Doubling time: ~207 days (~7 months)

Goodness of fit: $R^2 = 0.97$

Milestones: GPT-2 (~2s) → GPT-4 (~5m) → o3 (~1h50m)

Success Probability Drops Sharply as Task Length Increases for All Models



Interpretation

Each model has a downward success curve: longer tasks reduce completion probability. Frontier models shift curves rightward, preserving useful success at materially longer durations.

Reported 50% points: GPT-2 (~2s), GPT-4 (~5m), Claude 3.7 Sonnet (~54m), o3 (~1h50m). 80% time horizon follows the same pattern but is roughly 5× shorter.

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

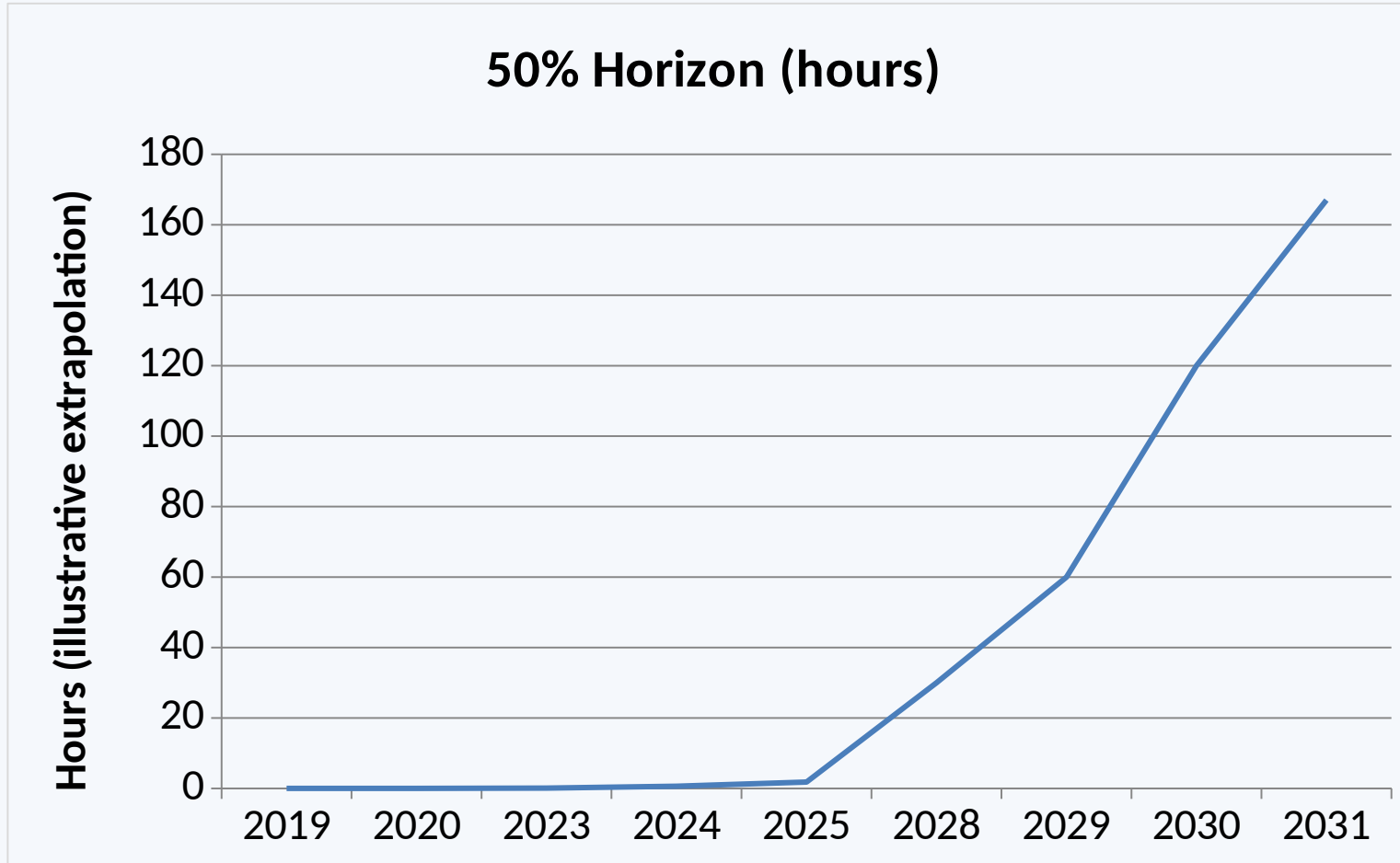
Observed capability drivers

- Improved logical reasoning over longer action chains
- More reliable tool use in coding and research workflows
- Greater self-awareness in execution and checking
- Better adaptation and recovery after mistakes

Limitations and open questions

- Performance is lower on less structured, “messier” tasks
- External validity remains uncertain across all real-world job distributions
- Benchmark tasks may not perfectly map to average intellectual labor segments

Extrapolation Predicts Month-Long Task Automation by 2028–2031



Forecast summary

Naive trend continuation implies ~167 work hours (~1 month) between mid-2028 and mid-2031.

2023–2025 growth appears ~20% faster than overall 2019–2025 trend.

Forecasts are conditional on unchanged scaling dynamics and benchmark relevance.

ys: Exponential Autonomous Capability Growth Demands Proactive Safety Gov

Policy and governance implications

- 1) The 50% time horizon is a robust, intuitive metric for tracking autonomy growth ($R^2 = 0.97$ over 6 years).
- 2) Time horizon doubling every ~7 months indicates sustained exponential capability gains, with signs of acceleration since 2024.
- 3) Frontier capability is now near 2-hour human-equivalent software tasks (o3), materially changing risk and deployment profiles.
- 4) External validity is unresolved: benchmark outcomes may not transfer uniformly across real-world intellectual labor.
- 5) Safety governance should emphasize eval cadence, capability thresholds, and controls for long-horizon autonomous operation.